Confluent Kafka Isotope

An example of using a Kafka **producer interceptor** to tag every message with a tracer — an *isotope* — that carries through every hop of a multi-topic pipeline. Consume-then-produce services adopt the inbound trace into thread-local with one explicit call (`IsotopeContext.adoptFromRecord(record)`) between consume and produce; the next `send()` then appends a fresh hop automatically. Apache Flink consumes the tagged records and reports on what the isotopes reveal: **end-to-end latency**, **hop topology**, **drop/duplication rates**, and **pipeline coverage**.

A **portability requirement** runs through this project: the same isotope mechanism must work against both **Confluent Cloud for Apache Flink (CCAF)** (managed) and **Confluent Platform for Apache Flink** (self-managed) — both used here via Table API SQL plus uploaded UDF/PTF JARs (no DataStream code on either side). Three decisions follow from that: tagging happens in a Kafka **producer interceptor** (the one extension point both runtimes share via the broker), the on-wire **header** format is **JSON** (so Flink SQL can read the scalar fields with `CAST(headers[…] AS STRING)` and no UDF), and the optional stateful reports (`LatencyPercentilesUDAF`, `StuckTracePTF`) ship as a single JAR that registers identically on either runtime.

One asymmetry the runtimes don't share: **Flink-native SR-Protobuf**. CCAF supports SR-framed Protobuf as a Flink sink format via its topic catalog; Apache Flink open-source (the CP Flink runtime) ships `avro-confluent` but no SR-Protobuf counterpart. So Flink *report sinks* land on **Avro+SR on CP** and can be **Protobuf+SR on CCAF** — a runtime constraint, not a project preference. The demo *event* topics (next paragraph) are unaffected because they're written by the Kafka producer client, not by Flink.

Message **values** on the demo topics are **SR-framed Protobuf** (`ai.signalroom.kafka.isotope.proto.DemoEvent`) — the standard Confluent value format. The interceptors and reports are agnostic to value format because the isotope rides in headers; the Protobuf choice just gives the integration tests and the demo CLI a typed payload to work with.

Table of Contents

1.0 How the isotope is carried

- **Header `x-isotope`** (JSON bytes) carries the full hop history, forwarded by every hop:
  - `t` — 16-byte **UUIDv7** trace ID (RFC 9562): 48-bit ms timestamp in the high bits + 74 bits random. Stable for the life of the trace,and lexicographic byte order matches creation order — sort trace IDs and you get chronological order for free.
  - `o` — origin timestamp (ms) — same value as the timestamp embedded in the UUIDv7 trace ID; kept as its own field for typed access from Flink SQL without needing to decode the UUID bytes.
  - `s` — origin service name (set once, never reassigned)
  - `h` — ordered list of hops, each `{s: service, t: topic, m: tsMs}`
  - `x` — `true` if the hop list exceeded `MAX_HOPS = 32` and the oldest hop was evicted
- **Six scalar headers** (UTF-8 strings) carry the most-recent-hop view so Flink SQL can read them via `CAST(headers['x-isotope-…'] AS STRING)` without parsing the JSON array (no UDF required on either CCAF or CP Flink). See scripts/flink/README.md for the full header table.

A producer with the isotope interceptor loaded appends one hop on every `send()`. A consume-then-produce service calls `IsotopeContext.adoptFromRecord(record)` between consume and produce so the trace ID and origin survive the hop.

How the producer interceptor gets invoked. Application code never calls `onSend` / `onAcknowledgement` directly — the Kafka client invokes them by reflection once two things are in place:

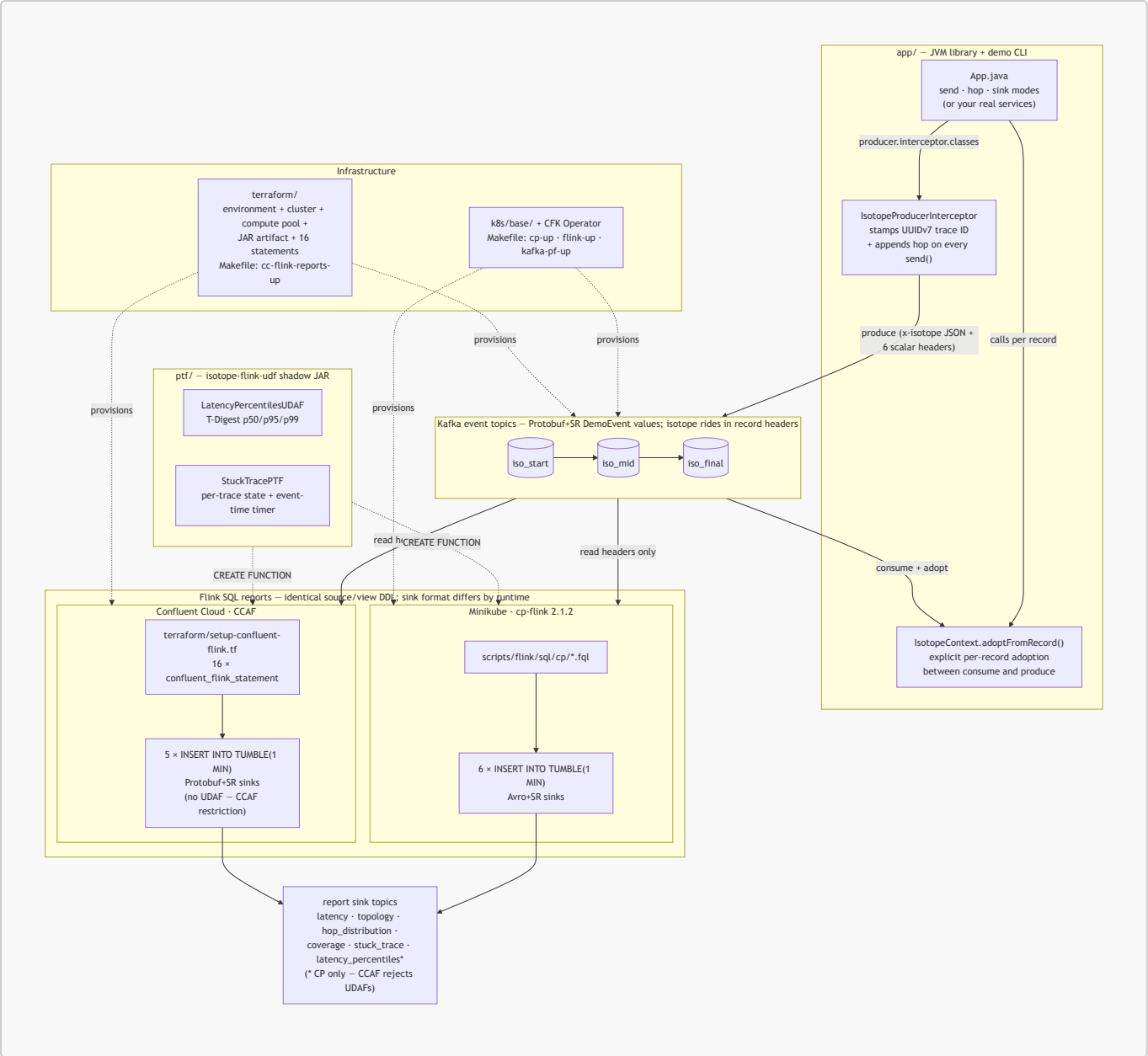
1. **Register the class in producer config.** Put `IsotopeProducerInterceptor.class.getName()` under `interceptor.classes` (App.java:185, App.java:250, IsotopeTestHarness.java:96). The `KafkaProducer` constructor instantiates the interceptor and owns its lifecycle.
2. **Call `producer.send(...)` as normal.** Every `send()` triggers `onSend` (caller thread, before serialization) and later `onAcknowledgement` (producer I/O thread, after broker ack or failure).

The exact call sites in `kafka-clients` 4.2.0: `KafkaProducer.send` invokes `interceptors.onSend` (line 950) and `AppendCallbacks.onCompletion` invokes `interceptors.onAcknowledgement` (line 1600). Exceptions thrown by the interceptor are caught and logged by the client's fan-out wrapper — they never break the `send` call.

Consumer-side adoption is explicit, not an interceptor. A `ConsumerInterceptor.onConsume` sees a whole batch from `poll()`, but the application processes records one at a time, so a thread-local snapshot from the batch would be ambiguous. Instead, services call `IsotopeContext.adoptFromRecord(record)` per record between consume and produce. The next `send()` then sees that thread-local and appends a new hop. This project ships no consumer interceptor at all — the producer interceptor plus per-record adoption is the entire mechanism.

2.0 Architecture

A bird's-eye view of the moving parts. The JVM library in app/ registers a Kafka producer interceptor that stamps the isotope into record headers on every `send()`; consume-then-produce services adopt the inbound trace via an explicit `IsotopeContext.adoptFromRecord(record)` call; records flow through a 3-topic chain; Flink SQL reads only the headers and emits 1-minute aggregate reports. The same source/view DDL deploys to both runtimes — **CP** on Minikube applies `.fql` files under scripts/flink/sql/cp/, and **CCAF** in Confluent Cloud applies inline `confluent_flink_statement` resources under terraform/. The Phase-2 shadow JAR from ptf/ registers identically on both. (Kafka is drawn once below for clarity — each runtime provisions its own cluster.)



See § 3.0 for the file tree behind each box, and § 4.0 for the run commands.

3.0 Repo layout

```
app/                                    isotope JVM library + demo CLI +
tests
  src/main/proto/ai/signalroom/kafka/isotope/proto/
    demo_event.proto                    DemoEvent message (Protobuf value
schema)
  src/main/java/ai/signalroom/kafka/isotope/
    Isotope.java                        POJO + JSON codec + Hop +
fromHeaders()
                                        + UUIDv7 helpers (uuidV7Bytes /
uuidV7String)
    IsotopeContext.java                 ThreadLocal + adoptFromRecord()
    IsotopeProducerInterceptor.java     stamps/appends x-isotope + 6 scalar
                                        reporting headers on send()
    App.java                            demo CLI — send / hop / sink modes
  src/test/java/.../                    IsotopeCodecTest (no broker needed)
```

```
  src/integrationTest/java/.../          BrokerSmokeIT,
ProducerInterceptorIT,
                                         ThreeStageHopPropagationIT,
                                         IsotopeTestHarness — live-broker
tests; produce/consume
                                         DemoEvent via SR-framed Protobuf
                                         (need Minikube CP + SR port-
forwarded)
ptf/                                     Phase 2 — Flink PTF + UDAF shadow
JAR
  src/main/java/ai/signalroom/kafka/isotope/flink/
    LatencyPercentilesUDAF.java          T-Digest p50/p95/p99 aggregate
    StuckTracePTF.java                   per-trace state + event-time timer
    Percentiles.java                     p50/p95/p99 return-type DTO
  src/test/java/.../                     LatencyPercentilesUDAFTest
k8s/base/                                CFK manifests
  confluent-platform-c3++.yaml           Kafka / SR / Connect / ksqlDB /
Control Center
  flink-basic-deployment.yaml            cp-flink session cluster + CMF
  flink-rbac.yaml                        RBAC for the cp-flink operator
scripts/
  port-forward-kafka.sh                  localhost:30092 → Kafka,
localhost:8081 → SR
  port-forward-taskmanager.sh            Flink TaskManager web UI forward
  deploy-cp-flink-reports.sh             builds shadow JAR + applies
sql/cp/*.fql to
                                         the cp-flink session cluster
  deploy-cc-flink-reports.sh             builds shadow JAR + wraps
`terraform apply`
                                         for the CCAF path
  cc-cli-env.sh                          pulls Kafka + SR creds from
`terraform output`,
                                         builds the JAAS string, exports
BOOTSTRAP /
                                         SR_URL / KAFKA_KEY / KAFKA_SECRET /
JAAS / ...
  cc-app-run.sh                          thin wrapper around `./gradlew
:app:run` that
                                         sources cc-cli-env.sh and injects
the six -D flags
  flink/README.md                        Flink SQL reports — runtime split
(CP=6 reports/Avro+SR,
                                         CCAF=5 reports/Protobuf+SR),
layout, operations
  flink/sql/cp/                          CP Flink SQL: 00_source_table,
01_register_functions,
                                         05_isotope_view, 05_report_sinks
(avro-confluent),
                                         10/20/30/40/60/70 INSERT INTO
reports, 99_teardown
                                         (CCAF SQL is inlined under
terraform/setup-confluent-flink.tf.)
terraform/                               CCAF infrastructure-as-code (`make
cc-flink-reports-up`)
```

```
  providers.tf                          Confluent provider — cloud
key/secret vars
  versions.tf                           required Terraform (>= 1.13) +
provider versions
  variables.tf                          confluent_api_key/secret, cloud,
region, day_count
  data.tf                               organization lookup + other data
sources
  setup-confluent-environment.tf        environment (ESSENTIALS stream-
governance package)
  setup-confluent-kafka.tf              Kafka cluster + Kafka API key
rotation module
                                        (iac-confluent-api_key_rotation-
tf_module)
  setup-confluent-flink.tf              service account + 6 role bindings,
compute pool,
                                        artifact upload, SR API key
rotation, and 16 inline
                                        `confluent_flink_statement`
resources: 3 ALTER TABLE
                                        + 2 VIEW + 5 sink CREATE TABLE + 1
CREATE FUNCTION
                                        (PTF; UDAF skipped — CCAF rejects
UDAFs) + 5 INSERT INTO
  outputs.tf                            environment_id, bootstrap, SR URL,
rotating
                                        Kafka + SR API key/secret outputs
(sensitive)
  terraform.png                         rendered resource graph (embedded
in § 4.5)
Makefile                                cp-up / flink-up / kafka-pf-up /
flink-reports-up /
                                        cc-flink-reports-up / cc-flink-
reports-down / ...
```

4.0 Running

4.1. Unit tests (no broker, instant)

```
./gradlew test                      # both subprojects
# or scoped:
./gradlew :app:test                 # IsotopeCodecTest           — JSON
roundtrip, hop eviction, header size, UUIDv7 properties (10 tests)
./gradlew :ptf:test                 # LatencyPercentilesUDAFTest — T-
Digest accumulator semantics
```

4.2 Demo CLI — see one trace propagate live

The fastest way to watch the isotope mechanic. Requires the cluster to be up and the Kafka + SR forwards running (see step 3 below for the bring-up commands). The CLI has three modes:

Mode	Args	What it does
send	<topic> <service> <payload>	Produces one isotope-tagged DemoEvent to <topic>, then exits. Auto-creates the topic.
hop	<in-topic> <out-topic> <service>	Consumes records from <in-topic>, adopts the isotope into thread-local, re-produces the same DemoEvent to <out-topic> as <service> (which appends a new hop). Runs until Ctrl-C.
sink	<topic>	Subscribes to <topic> and pretty-prints the full isotope trail for every arriving record. Runs until Ctrl-C.

A 3-stage chain in four terminals:

```
# Terminal A — terminal sink (will print the full 3-hop trail)
./gradlew :app:run --args="sink iso_final" -q

# Terminal B — middle stage: iso_mid → iso_final as svc-C
./gradlew :app:run --args="hop iso_mid iso_final svc-C" -q

# Terminal C — first stage: iso_start → iso_mid as svc-B
./gradlew :app:run --args="hop iso_start iso_mid svc-B" -q

# Terminal D — kick the chain off (run repeatedly to send more)
./gradlew :app:run --args="send iso_start svc-A 'hello world'" -q
```

Terminal A's output for each record shows the same trace_id across all three hops, origin = svc-A (never reassigned), and hops[] listing svc-A → svc-B → svc-C in order with per-hop timestamps. Override endpoints via -Dkafka.bootstrap=… / -Dschema.registry.url=… if you're not on the default Minikube layout.

4.3 Integration tests (live Kafka via Minikube)

Bring up the local Confluent Platform stack and port-forward Kafka + SR:

```
make minikube-start                  # one-time
make cp-up                           # CFK Operator +
Kafka/SR/Connect/ksqlDB/C3 (~5 min)
make kafka-pf-up                     # localhost:30092 → Kafka,
localhost:8081 → Schema Registry
```

Then run the suite:

```
./gradlew :app:integrationTest                                          #
every IT below
```

```
./gradlew :app:integrationTest --tests '*ProducerInterceptorIT'          #
just one
```

Override the endpoints if needed:

```
./gradlew :app:integrationTest \
    -PkafkaBootstrap=localhost:30092 \
    -PschemaRegistryUrl=http://localhost:8081
```

Tear down forwards when done:

```
make kafka-pf-down
```

The integration tests cover:

Test	What it verifies
`BrokerSmokeIT`	AdminClient can create/list/delete a topic via the NodePort port-forward
`ProducerInterceptorIT`	A consumer sees the `x-isotope` JSON header + all 6 scalar reporting headers with the expected origin/hop values, and the Protobuf round-trip preserves `DemoEvent.source` / `payload`
`ThreeStageHopPropagationIT`	`svc-A → topic-AB → svc-B → topic-BC → svc-C` produces a stable trace ID, 2-hop trail in send order, and correct scalar headers (origin = `svc-A`, this = `svc-B`, hop count = 2) at the terminal; consume-then-produce hops use `IsotopeContext.adoptFromRecord` to carry the trace forward

4.4 Flink SQL reports on Confluent Platform for Apache Flink (Minikube)

The four Phase-1 reports plus the Phase-2 PTF and UDAF reports — six in total — run against a Flink session cluster managed by the Confluent Flink Kubernetes Operator. Same FQL files deploy to Confluent Cloud for Apache Flink — see **§ 4.5** for that path; this section is the local-Minikube path.

Bring up Flink:

```
make flink-up                 # cert-manager → CFK Flink Operator → CMF →
session cluster
                              # (~5 min the first time)
```

Deploy the reports — all 6 reports run on the cp-flink session cluster (Flink 2.1.2). Sink topics use Apache Flink's `avro-confluent` format — SR-framed Avro, auto-registered on first write — so Control Center renders the report rows natively.

```
make flink-reports-up
```

`flink-reports-up` builds the PTF/UDAF shadow JAR if missing, copies it into the JobManager pod, pre-creates the 6 sink Kafka topics, then applies the source + view + sink DDL and submits 6 `INSERT INTO` streaming jobs (one per report). On first write to each sink, Apache Flink's `flink-sql-avro-confluent-registry` format registers a fresh Avro schema in SR under subject `<topic>-value`. **Control Center deserializes all 6 report topics natively** — no `.proto` files in the repo for the reports, no hand-installed format jars beyond the one init-container download.

4.4.1 Format-by-domain

The demo *event* topics (`iso_start`, `iso_mid`, `iso_final`) still ride **Protobuf+SR** via the Java app's `DemoEvent` schema — that's unchanged. The *report* topics ride **Avro+SR** because cp-flink doesn't ship an SR-integrated Protobuf format and CMF (which does) disallows the UDAFs the percentiles report needs. Events from the app are Protobuf; aggregates from Flink are Avro. Two formats by domain — a clean split, not a defect.

4.5 Flink SQL reports on Confluent Cloud for Apache Flink (CCAF)

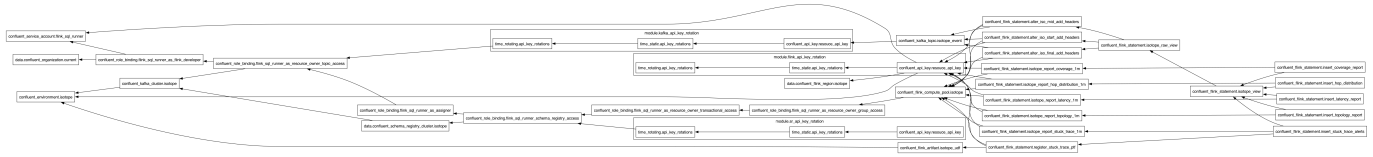
CCAF parallel of § 4.4, driven by Terraform under terraform/. One `make` target spins up a fresh Confluent Cloud environment, Kafka cluster, 3 pre-created isotope event topics (report sink topics are created on the fly by the Flink `CREATE TABLE` statements — see the comment in terraform/setup-confluent-kafka.tf for why pre-creating them via `confluent_kafka_topic` would conflict with CCAF's Topic Catalog auto-import), a Flink compute pool, two rotating service-account API key pairs (one for Kafka, one for Schema Registry), the PTF/UDAF JAR uploaded as a Flink artifact, and 16 long-lived `confluent_flink_statement` resources broken down as **3 ALTER TABLE** (add scalar headers on the event topics) + **2 VIEW** (raw + typed) + **5 sink CREATE TABLE** + **1 CREATE FUNCTION** (PTF only — the UDAF is intentionally skipped, see the limitation note below) + **5 INSERT INTO** streaming jobs. The Terraform shape mirrors `apache_flink-kickstarter-ii` — same provider version, same `iac-confluent-api_key_rotation-tf_module`, same DROP-then-CREATE statement pattern.

Prereqs:

- Terraform `>= 1.13` installed locally.
- A Confluent Cloud API key (Cloud-level, not cluster-scoped) with permissions to create environments, Kafka clusters, Flink compute pools, service accounts, role bindings, Flink artifacts, and statements. Generate via Console → Settings → Cloud API keys.

Deploy:

```
export CONFLUENT_API_KEY=...
export CONFLUENT_API_SECRET=...
make cc-flink-reports-up CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```



The wrapper script (scripts/deploy-cc-flink-reports.sh) builds the PTF/UDAF shadow JAR if missing, then runs `terraform apply -auto-approve` in terraform/. First-run takes ~6–8 minutes (Kafka cluster provisioning dominates). Re-applies are idempotent — `CREATE … IF NOT EXISTS` plus `lifecycle { ignore_changes = [compute_pool] }` on every statement.

What gets created (see terraform/setup-confluent-flink.tf for the full graph):

Resource	Name	Notes
`confluent_environment`	`confluent-kafka-isotope`	ESSENTIALS stream-governance package
`confluent_kafka_cluster`	`kafka-isotope`	Standard, single-zone, AWS us-east-1 by default
`confluent_kafka_topic` × 3	`iso_{start,mid,final}`	Only the event topics are pre-created. The 5 `isotope_report_*_1m` sink topics are created on first deploy by their `CREATE TABLE` statement (CCAF's Topic Catalog auto-imports any pre-existing topic as `(key BYTES, val BYTES)`, which would silently no-op the typed `CREATE TABLE`). `terraform destroy` cleans them up via the environment cascade.
`confluent_service_account` + 6 role bindings	`isotope-flink-sql-runner`	FlinkDeveloper (org) + ResourceOwner on topic=* / transactional-id=* / group=* / SR subject=* + Assigner on the service account
`confluent_flink_compute_pool`	`isotope-flink-statement-runner`	10 CFU; headroom for 5 INSERTs + ad-hoc SELECTs
`confluent_flink_artifact`	`isotope-flink-udf`	Uploads `ptf/build/libs/isotope-flink-udf.jar`
`confluent_flink_statement` × 16	(see file)	3 ALTER TABLE (event-topic scalar headers) + 2 VIEW (raw + typed) + 5 sink CREATE TABLE + 1 CREATE FUNCTION (PTF

Resource	Name	Notes
		only; UDAF skipped) + 5 INSERT INTO

Useful outputs:

```
terraform -chdir=terraform output environment_id
terraform -chdir=terraform output kafka_bootstrap_servers
terraform -chdir=terraform output schema_registry_url
terraform -chdir=terraform output -raw kafka_api_key     # sensitive
terraform -chdir=terraform output -raw kafka_api_secret  # sensitive
```

Format-by-runtime (not -by-domain). CP's reports land on **Avro+SR** (`'value.format' = 'avro-confluent'` in scripts/flink/sql/cp/05_report_sinks.fql). CCAF's reports land on **Protobuf+SR** (`'value.format' = 'proto-registry'` in each sink's WITH clause in terraform/setup-confluent-flink.tf). The two runtimes' SQL is otherwise unshared: CP's lives hardcoded in scripts/flink/sql/cp/, CCAF's lives inline as `confluent_flink_statement` resources in terraform/setup-confluent-flink.tf.

CCAF UDAF limitation. CCAF currently rejects all `CREATE FUNCTION` statements for user-defined aggregate functions ("aggregate functions are not supported"). The `LATENCY_PERCENTILES` UDAF — Phase-2 of the project — therefore deploys on CP only; the percentile report (`latency_percentiles_flat_1m`) does not exist on CCAF. The other Phase-2 function, `STUCK_TRACE_PTF` (a ProcessTableFunction, not a UDAF), works on both runtimes. The JAR itself is portable — `LatencyPercentilesUDAF` ships with a byte[]-based accumulator and is ready to register when Confluent lifts the restriction. The five CCAF reports are: `latency` (avg/min/max), `topology`, `hop_distribution`, `coverage`, `stuck_trace`.

Driving traffic — the 3-stage demo against CCAF. App.java reads four optional `-D` properties (`kafka.security.protocol`, `kafka.sasl.mechanism`, `kafka.sasl.jaas.config`, `schema.registry.basic.auth.user.info`) that default to plaintext-no-auth for Minikube. scripts/cc-cli-env.sh pulls the Kafka + Schema-Registry credentials from `terraform output` (both keys are rotated by `module.kafka_api_key_rotation` and `module.sr_api_key_rotation` in terraform/setup-confluent-kafka.tf) and builds the JAAS string.

The thin wrapper scripts/cc-app-run.sh sources the env helper then invokes `./gradlew :app:run` with the six `-D` flags — pass it the same `send` / `hop` / `sink` args you'd give App.java:

```
# Four terminals (same A/B/C/D order as § 4.2). No manual env exports —
# the wrapper sources cc-cli-env.sh, which pulls everything from terraform.
scripts/cc-app-run.sh sink iso_final                # A
scripts/cc-app-run.sh hop iso_mid iso_final svc-C   # B
scripts/cc-app-run.sh hop iso_start iso_mid svc-B   # C
scripts/cc-app-run.sh send iso_start svc-A 'hello'  # D
```

The wrapper hard-fails with a clear message if any of the seven required values is missing, so you'll never silently hand gradle empty `-D` values.

Terminal A prints the same `trace_id` across all three hops, and the CCAF report INSERTs populate as you fire Terminal D — `SELECT * FROM isotope_report_latency_1m` etc. in the Cloud Console SQL workspace.

Sustained traffic — required to see report rows. The five INSERT INTO jobs aggregate over `TUMBLE(event_time, INTERVAL '1' MINUTE)` windows, and a tumbling window only emits when the watermark advances past `window_end`. A handful of records bursted from Terminal D within a single 1-minute interval will sit in one open window forever (the most-recent record is the watermark, and it never gets older than itself). Spread traffic across **multiple** windows so the watermark crosses each boundary:

```
# 30 records spaced 5s apart ≈ 2.5 minutes of event-time → spans 3+ windows
for i in {1..30}; do
  scripts/cc-app-run.sh send iso_start svc-A "burst-$i"
  sleep 5
done
```

Wait ~90 seconds after the *last* record before checking `isotope_report_latency_1m` (and friends) — that's the watermark catching up. The `stuck_trace_alerts_1m` sink only fires for traces that go ≥60s of event time without a fresh hop, so the burst above won't trigger it (every trace gets one record and ends — no stalled in-flight state). To exercise `STUCK_TRACE_PTF`: send a single record to `iso_start` and don't run the `svc-B` / `svc-C` hops, then keep sending unrelated records elsewhere so the watermark advances past the stuck trace's `event_time + 60s`.

> **Tip — auto-source after apply.** `make cc-flink-reports-up` runs in its own subshell, so it can't export env vars back into yours. Add this to your `~/.zshrc` / `~/.bashrc` for a one-liner that applies and exports:
>
> ```
> cc-up() {
>   make cc-flink-reports-up "$@" && source scripts/cc-cli-env.sh
> }
> cc-down() {
>   make cc-flink-reports-down "$@" && unset BOOTSTRAP SR_URL KAFKA_KEY
> KAFKA_SECRET SR_KEY SR_SECRET JAAS
> }
> # then:  cc-up CONFLUENT_API_KEY=... CONFLUENT_API_SECRET=...
> ```

Teardown:

```
make cc-flink-reports-down CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```

Runs `terraform destroy -auto-approve` — deletes every resource above, including the environment itself. Safe to run repeatedly.

4.6 Recommended path the first time through

1. `./gradlew test` — proves the codec + UDAF logic without any cluster.
2. `make cp-up && make kafka-pf-up && ./gradlew :app:integrationTest` — proves the broker + SR + interceptor + Protobuf path end-to-end.
3. The 3-stage demo CLI walkthrough above — visually shows the trace accumulating hops.
4. `make flink-up && make flink-reports-up && make flink-sql` — reports populate as you drive traffic via the demo CLI (see § 4.2).
5. (Optional) `make cc-flink-reports-up` — the CCAF parallel; see § 4.5 for prereqs and the SASL-config caveat for the demo CLI.